

联系

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教育

- 2022—2026
清华大学
工程物理系 · 本科
- 即将入学
核工业西南物理研究院
核能科学与技术 · 硕士

工具

- Python
- MATLAB
- 数据分析
- 物理建模
- 学术写作

探索聚变，记录思考。

清华大学工程物理系核工程与核技术专业本科生，即将前往核工业西南物理研究院攻读核能科学与技术方向硕士。研究兴趣集中于磁约束核聚变、托卡马克边缘等离子体与高 Z 杂质辐射。

研究经历

2025—2026 · 本科综合论文训练

托卡马克等离子体 EUV 辐射的研究

清华大学工程物理系 · 指导教师：谭熠 副教授

核心问题：托卡马克能否作为 13.5 nm 工业级极紫外光刻光源？

≈ 0.5%

等离子体层 CE 中值

10⁻⁴—10⁻³

系统层有效 CE

≈ 10⁸×

Etendue 失配

- 建立输入功率—总辐射—Sn/Xe 杂质辐射—13.5 nm 带内输出的分层转换效率模型。
- 从参数匹配、能量效率、辐射亮度、MHD 稳定性与工程集成等维度对比托卡马克和 LPP 光源。
- 结论表明托卡马克难以成为工业 EUV 光源，但可作为高 Z 杂质辐射与边缘输运研究平台。

研究方向

- 磁约束核聚变
- 托卡马克
- 偏滤器物理
- 高 Z 杂质输运
- EUV 辐射
- 碰撞辐射模型

Jingchen Liu

Nuclear Engineering
Magnetic Confinement
Fusion
Plasma Physics

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EDUCATION

- 2022—2026
Tsinghua University
Engineering Physics · B.Eng.
- Incoming
Southwestern Institute of Physics
Nuclear Science & Technology

TOOLS

- Python
- MATLAB
- Data Analysis
- Physical Modeling
- Academic Writing

Exploring fusion, documenting ideas.

Undergraduate in Nuclear Engineering and Nuclear Technology at Tsinghua University. Incoming graduate student at the Southwestern Institute of Physics, focusing on magnetic confinement fusion, tokamak edge plasmas, and high-Z impurity radiation.

RESEARCH EXPERIENCE

2025—2026 · UNDERGRADUATE THESIS

Feasibility of Tokamak Plasma as an EUV Radiation Source

Department of Engineering Physics, Tsinghua University · Supervisor: Assoc. Prof. Yi Tan

Research question: Can a tokamak serve as an industrial 13.5 nm EUV lithography source?

≈ 0.5%

Median plasma-level CE

10⁴—10³

System-level CE

≈ 10⁸×

Etendue mismatch

- Built a layered conversion-efficiency model from input power to total radiation, Sn/Xe impurity radiation, and 13.5 nm in-band output.
- Compared tokamak and LPP sources across parameter matching, efficiency, radiance, MHD stability, and engineering integration.
- Concluded that tokamaks are unlikely industrial EUV sources but remain valuable platforms for high-Z impurity and edge-transport research.

RESEARCH FOCUS

Magnetic Confinement

Tokamak

Divertor Physics

High-Z Impurities

EUV Radiation

C-R Modeling